

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Please provide a brief explanation of your answer.

Thought for 98 seconds >

$$x_1 = 0, x_2 = 1$$

The total dosage constraint is $1 \cdot x_1 + a_2 \cdot x_2 = 1$, with $x_1, x_2 \geq 0$. The objective is to minimise total time $T = x_1 + x_2$. Substituting $x_1 = 1 - a_2 x_2$ gives $T = 1 + (1 - a_2)x_2$.

- If $a_2 > 1$, T is minimised by making x_2 as large as possible: $x_2 = 1/a_2$, $x_1 = 0$.
- If $a_2 < 1$, T is minimised by $x_2 = 0$, $x_1 = 1$.
- If $a_2 = 1$, any nonnegative x_1, x_2 with sum 1 is optimal.

The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$ as $n \rightarrow \infty$. Evaluating $f(n)$ yields a sum whose limit is

$$a_2 = 1 + 2 \cdot 72^{-M} - \frac{2 \cdot 72^{-3M}}{1 - 72^{-M}},$$

with $M = 10^{24}$. The correction term is positive (since $72^{-M} \gg 72^{-3M}$) but astronomically small. Hence $a_2 = 1 + \varepsilon$ with $\varepsilon > 0$ unimaginably tiny. The strict inequality $a_2 > 1$ makes the unique optimal solution $x_1 = 0$, $x_2 = 1/a_2$.

Approximating $1/a_2 \approx 1$ gives an ℓ^∞ error well below 0.005, satisfying the two-digit accuracy requirement.